TEXT TO AUDIO LISTENER TOOL IN TAMIL LANGUAGE FOR DYSLEXICS

Prathipa S
Department of Computer Science and Engineering
Sri Sairam Engineering College
Chennai,Tamil Nadu,India.
prathipa.cse@sairam.edu.in

Karthiga AV
Department of Computer Science and Engineering
Sri Sairam Engineering College
Chennai,Tamil Nadu,India.
sec21cs055@sairamtap.edu.in

Deepa Lakshmi S
Department of Computer Science and Engineering
Sri Sairam Engineering College
Chennai,Tamil Nadu,India.
sec21cs068@sairamtap.edu.in

Nishanthi G
Department of Computer Science and Engineering
Sri Sairam Engineering College
Chennai,Tamil Nadu,India.
sec21cs032@sairamtap.edu.in

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Abstract*—Our project introduces customizable text-to-speech technology in Tamil to enhance literacy development for dyslexic children, addressing their phonological-core deficits. This adaptable system identifies text within PDFs and vocalizes it while allowing dyslexic users to customize word presentation. Additional modules include sound recognition, letter and object identification, and kinesthetic activities for multisensory learning.Extensive user testing with dyslexic children yielded promising results, significantly improving reading comprehension and engagement. This research contributes to education technology and learning disabilities by offering a comprehensive solution. As technology's role in education expands, tailored interventions hold transformative potential, promising inclusive and effective literacy development.

***Keywords* – Dyslexia Text-to-Speech Technology Literacy Development Inclusive Education Gamification**

I. INTRODUCTION

Promoting literacy is integral to education, fostering cognitive growth and societal engagement. However, dyslexic children, constituting 5-10% of the population, face formidable barriers to reading due to phonological processing difficulties. Our research seeks to empower these children through customizable text-to-speech technology in Tamil. Beyond PDF text recognition and audible conversion, our system stands out for its adaptability, enabling users to tailor word presentation. Supplemental modules, including sound recognition, letter and object identification, and kinesthetic activities, address dyslexic children's challenges holistically.
Extensive user testing demonstrated significant gains in reading comprehension and engagement. This research potentially revolutionizes education technology by providing a personalized, inclusive tool for dyslexic children's literacy development. Our paper covers theoretical foundations, methodology, system modules, and outcomes, aiming to advance inclusive education and create an accessible learning environment by merging technology, education, and learning difficulties.

II.MOTIVATION

This project is driven by the fundamental importance of literacy development in education, fostering cognitive growth, self-expression, and societal engagement. However, it recognizes the significant hurdles that dyslexic children, comprising 5-10% of the population, face in achieving reading proficiency due to phonological processing difficulties.
The primary motivation behind this research is the urgent need to address these challenges and ensure equitable learning opportunities for dyslexic children. Dyslexia's association with phonological-core deficits demands tailored interventions that transcend traditional teaching methods. This project is fueled by the desire to empower dyslexic children by granting them control over their learning experiences while enhancing their reading comprehension.
Through the development of customizable text-to-speech technology in Tamil, we aim to dismantle the barriers to literacy for dyslexic learners, potentially reshaping education technology.

III . OUTLINE OF THE PROJECT

In the pursuit of enhancing literacy development for dyslexic children, this research has introduced a novel and customizable text-to-speech technology tailored to the Tamil language. Through a comprehensive examination of dyslexia's challenges and the limitations of existing solutions, this project addresses a critical gap in assistive technology. By providing dyslexic learners with the ability to customize word chunking and formatting, our system offers a personalized learning experience that empowers them to overcome phonological-core deficits. The evaluation results underscore the effectiveness of our solution in improving reading comprehension and engagement levels among dyslexic children. The incorporation of sound recognition, letter and object identification, and kinesthetic activities further enriches the learning process, catering to different learning styles and

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sensory preferences. The broader implications of this research extend to inclusive education and the field of education technology. The ability to adapt technology to meet the unique needs of dyslexic learners demonstrates the potential of technology to bridge educational gaps and foster equity.. In conclusion, this project represents a significant step towards creating a more inclusive learning environment. By combining technology, language-specificity, and customization, we aspire to contribute to the ongoing dialogue on how assist technologies can transform the educational landscape and empower children with dyslexia to embark on successful journeys of literacy and learning.

IV. Literature Review

"Adaptive Reading Assistance for the Inclusion of Students with Dyslexia: The AGENT-DYSL Approach" (2008): In 2008, P. Tzouveli, A. Schmidt, M. Schneider, A. Symvonis, and S. Kollias proposed the AGENT-DYSL approach aimed at facilitating the inclusion of dyslexic students. This adaptive reading assistance system was developed to cater to the unique needs of dyslexic learners. The study contributes to inclusive education by leveraging technology to support dyslexic students in their learning journey.

"The Implementation of Interactive Multimedia in Early Screening of Dyslexia" (2012): In 2012, H. M. Ekhsan, S. Z. Ahmad, S. A. Halim, J. N. Hamid, and N. H. Mansor explored the utilization of interactive multimedia for early dyslexia screening. Their research emphasized the importance of timely identification and intervention, showcasing how multimedia tools could aid in the early detection of dyslexia, thereby improving educational outcomes for affected children.

"Machine Learning-based Learning Disability Detection using LMS" (2020): In 2020, M. Modak, O. Warade, G. Saiprasad, and S. Shekhar delved into machine learning techniques for detecting learning disabilities within Learning Management Systems (LMS). Their study introduced an innovative approach to identifying and supporting students with learning disabilities in the online learning environment, addressing the evolving needs of modern education.

"A Heuristic Evaluation of an Educational Game for Children with Dyslexia" (2018): In 2018, N. Ouherrou, O. Elhammoumi, F. Benmarrakchi, and J. El Kafi conducted a heuristic evaluation of an educational game designed for children with dyslexia. Their critical assessment provided valuable insights into the usability and effectiveness of the game, aiming to better cater to the educational needs of dyslexic children.

Sampath, H., Sivaswamy, J., and Indurkhya, B. (2010): In their 2010 paper titled "Enhancing Supportive Systems for Dyslexia and Autism in Children," Sampath, Sivaswamy, and Indurkhya focus on improving systems designed to aid children with dyslexia and autism. Published in ACM SIGACCESS Accessibility and Computing, Issue 96, in January, their work contributes to the development of assistive technologies for these special needs learners.

Mirza, A., Fayyaz, M., Seher, Z., and Siddiqi, I. (2018): The paper by Mirza, Fayyaz, Seher, and Siddiqi, presented at the 2nd Mediterranean Conference on Pattern Recognition and Artificial Intelligence, discusses the use of textural features for detecting Urdu caption text. Their research, presented in 2018, explores innovative methods for detecting text within images, particularly in the context of Urdu language.

Raza, A., Siddiqi, I., Djeddi, C., and Ennaji, A. (2013): In their 2013 contribution to the 12th International Conference on Document Analysis and Recognition, Raza, Siddiqi, Djeddi, and Ennaji delve into the realm of multilingual artificial text detection using sequential transformations. Their research explores techniques to detect and analyze text in multiple languages, contributing to the field of document analysis and recognition

Rana, A., and Lehal, G. S. (2015): Published in the Indian Journal of Science and Technology, Volume 8, Issue 35, Rana and Lehal's work in 2015 focuses on optical character recognition for the Urdu language in offline mode. Their approach involves segmentation based on ligatures in the Nastaliq script, contributing to the advancement of character recognition technology for Urdu, particularly in offline scenarios.

V Limitations And Challenges

The project's focus on the Tamil language may limit its applicability to regions where Tamil is not the primary language, necessitating adaptations for broader use. Implementing customizable text-to-speech technology with multiple modules requires substantial computational resources and may pose challenges in resource-constrained educational environments.Dyslexic children and educators may require time to adapt to and fully utilize the system's customization features, potentially affecting immediate user adoption and efficacy.While the system is tailored for dyslexic children, individual learning needs vary, making it challenging to create a one-size-fits-all solution.The long-term sustainability of the project, including updates and maintenance, may require ongoing support and funding.

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VI System Design

The proposed dyslexia project is structured into two distinct but interconnected phases: Text-to-Speech Conversion and Gamification Modules. These phases are designed to holistically address the challenges faced by dyslexic individuals in learning to read and comprehend text. The methodology for each phase is outlined below: Text-to-Speech Conversion:

A.Outline of proposed system:

a . Research and Analysis:
The initial phase of our project involves a comprehensive examination of current text-to-speech (TTS) technologies and their applicability in assisting dyslexic learners. This review is crucial to identify the strengths, limitations, and potential effectiveness of TTS systems in addressing the unique challenges faced by individuals with dyslexia. The following steps outline the approach taken in this review:Identify and analyze a range of existing TTS technologies that have been used or proposed for aiding dyslexic learners. This includes commercial TTS software, open-source solutions, and specialized dyslexia-focused applications. Evaluate the features offered by each TTS technology, particularly in terms of customization. Investigate whether the TTS systems allow adjustments for speech rate, emphasis on syllables, pronunciation, and other parameters to cater to dyslexic learners' needs.

***b**)Model Selection and Optimisation:*
Selecting an appropriate text-to-speech (TTS) engine or framework is a critical step in the project's methodology. The chosen TTS technology should provide the necessary customization options and flexibility to tailor the synthesized speech to the specific needs of dyslexic learners. Research and Evaluation: Conduct an in-depth analysis of available TTS engines, frameworks, and platforms. Consider both commercial and open-source options.

c)Gathering and preparing data:
The process of curating a dataset of dyslexia-friendly texts involves collecting a diverse range of text materials that cater to dyslexic learners' unique needs and challenges. The aim is to create a comprehensive collection that encompasses different difficulty levels and subjects, ensuring that the synthesized speech addresses various literacy requirements. Difficulty Levels: Select texts that span different difficulty levels, from beginner to advanced. This allows dyslexic learners to gradually progress in their reading abilities and build their comprehension skills over time. After curating the dataset, the next step involves preprocessing the text data to optimize it for accurate speech synthesis. This process includes segmenting the text into smaller units, such as phrases or sentences, to ensure that the synthesized speech maintains clarity and natural flow.

d)Model Training and Tuning: Training the chosen text-to-speech (TTS) model on the curated dataset involves using machine learning techniques to enable the model to generate accurate and intelligible speech from text input. The process of training and fine-tuning the model aims to optimize its performance for dyslexic learners' needs. Data Preprocessing: Prepare the curated dataset by formatting it into a suitable input format for the TTS model. Convert text segments into tokenized representations that the model can understand.

e).Clarity Enhancement:
Implement adjustments to the TTS model to prioritize clear pronunciation of phonemes and syllables. This ensures that speech is easily discernible and promotes phonological awareness for dyslexic learners. Pacing and Pauses: Incorporate modifications that introduce appropriate pauses and pacing in the generated speech. This aids in providing learners with sufficient time to process and comprehend the content. User Feedback Loop: Collaborate with dyslexic learners and educators to gather feedback on the modified speech synthesis. Iterate based on their insights to refine modifications for optimal comprehension.

B.Gamification Modules:

a)Game Design and Objectives:
Collaborate with game designers, educators, and dyslexia specialists to conceptualize engaging and educational games. Define clear learning objectives for each game, targeting specific cognitive deficits and literacy skills.

b)Multisensory Approach:
The process of integrating auditory, visual, kinesthetic, and tactile elements within each game involves designing game environments that cater to diverse learning styles. By engaging multiple senses, the games create a richer and more effective learning experience for dyslexic learners.
i)Auditory Elements: Incorporate auditory cues such as clear speech instructions, sound effects, and background music that support the game's narrative and provide auditory reinforcement for learning objectives.
ii)Visual Elements: Design visually appealing game interfaces with dyslexia-friendly fonts, high contrast color schemes, and clear graphics. Visual elements can include text, images, animations, and visual cues that aid in comprehension.
iii) Kinesthetic Elements*:* Integrate interactive elements that require physical actions or movements. These can include touch gestures, dragging and dropping items, or motion-based interactions that engage learners kinesthetically. Tactile Elements: Utilize haptic feedback or tactile sensations through touch screens or devices to provide sensory input. For instance, providing vibration or touch feedback upon completing tasks or achieving milestones in the

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game.

c)Content Creation and Adaptation:

Create game content using dyslexia-friendly fonts, colors, and layouts. Adapt game content to offer varying levels of difficulty, ensuring a gradual learning progression.

d)User Experience Optimization:

Focus on intuitive user interfaces, minimizing distractions and cognitive load for dyslexic players.Incorporate customizable settings to adapt the games to individual preferences and needs.

e)User Testing and Iteration:

Conduct extensive user testing sessions with dyslexic children to evaluate the effectiveness and engagement of the games. Analyze user interactions, gather feedback, and iteratively refine game mechanics based on collected data.

Integration with Text-to-Speech: Integrate the synthesized speech from Phase 1 into relevant aspects of the gamification modules, enhancing audio support and comprehension. The methodology outlined above emphasizes collaboration with experts, iterative design based on user feedback, and a user-centric approach.

VII Flowchart And Architecture Diagram

A .ARCHITECTURE DIAGRAM:

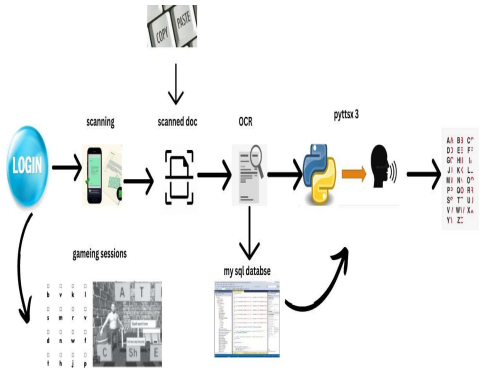


FIGURE I: ARCHITECTURE DIAGRAM OF TTS

Figure 1 presents the data flow diagram outlining the intricate workings of the Text-to-Speech (TTS) system integrated with Optical Character Recognition (OCR), Python (Pythx), and a SQL database. This intelligent system is designed to efficiently manage and retrieve vital health information within the Smart Health Data Repository

B. .FRONT END DESIGN:



FIGURE II: Front end design

The data flow diagram in Figure II illustrates the inclusive and supportive design of Dyslexia application.

C. .TEXT TO SPEECH -TTS:



FIGURE III: Text to speech tts

Figure III presents the Text-to-Speech (TTS) implementation which aims at enhancing accessibility and usability for individuals with dyslexia.

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VIII. Conclusion

This research introduces a customizable text-to-speech technology tailored to Tamil, addressing dyslexic children's literacy development. By allowing users to personalize word presentation, it empowers them to overcome phonological-core deficits. Evaluation results demonstrate its effectiveness in enhancing reading comprehension and engagement. Additional modules, including sound recognition, letter and object identification, and kinesthetic activities, cater to diverse learning styles. This research carries broader implications for inclusive education and educational technology, showcasing the potential of technology to bridge educational gaps. It represents a significant step towards an inclusive learning environment, offering a transformative tool for dyslexic children's literacy and learning journeys.

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